Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 1 of 7 · Question 4 part b.iii · 4 marks

2025 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

4	(a)	(i)	• decreases; • from 82% to 72%/by 10%		2
		(ii)	79:21; 3.76:1;	accept 3.7612:1 accept 3.762:1 accept 3.8/4:1	2
		(iii)	any two from • passive smoking/ex- smokers already have damaged lungs; • (air) pollution; • genetics/heredity;		2
		(iv)	tar;		1
	(b)	(i)	bronchiole;		1
		(ii)	alveolus;		1
		(iii)	any four from • damaged alveoli/alveoli walls broken; • less surface area (for gas exchange); • difficulty breathing/shortness of breath; • less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange; • reduced (aerobic) respiration; • fatigue/less energy/not very active;		4
				Total for Question 4 = 13 marks	

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 2 of 7 · Question 6 part c.ii · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

6	(a) (i)	Increase/goes up;	allow equivalent	1
	(ii)	Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;;	Allow one mark for reference to lifestyles only	2
	(b)	Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed; Increased risk of (lung/respiratory) infection/coughing;		1 1
	(c) (i)	<table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table>	Allow for SA:V ratio 6:3 ECF	3
	(ii)	Smaller surface area to volume ratio (in lungs with emphysema); Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide; Less oxygenated blood/less oxygen transferred (to cells); Less (aerobic) respiration;		Max 3

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 2 of 7 · Question 6 part c.ii · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 3 of 7 · Question 3 part a.ii · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a) (i)	C; (red blood cells)		1
(ii)	any three from • more carbon dioxide (1) • less oxygen (1) • moister (1) • warmer (1)		3
(iii)	• large surface area (1) • thin (walled)/wall one cell thick (1) • good blood supply (1) • moist lining (1)		3
(b)	• thin wall (1) • narrow lumen (1)	Allow references to only one blood cell passing through	2
(c)	• breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1) • as lungs would not squeeze out air/forced exhalation(1)		2

Total 11 marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 3 of 7 · Question 3 part a.ii · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 4 of 7 · Question 3 part a.iii · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

3 (a) (i)	C; (red blood cells)		1
(ii)	any three from • more carbon dioxide (1) • less oxygen (1) • moister (1) • warmer (1)		3
(iii)	• large surface area (1) • thin (walled)/wall one cell thick (1) • good blood supply (1) • moist lining (1)		3
(b)	• thin wall (1) • narrow lumen (1)	Allow references to only one blood cell passing through	2
(c)	• breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1) • as lungs would not squeeze out air/forced exhalation(1)		2

Total 11 marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 4 of 7 · Question 3 part a.iii · 3 marks

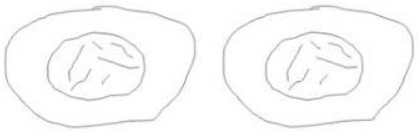
2020 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 5 of 7 · Question 2 part a.iii · 4 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a) (i)	156 – 72; 84;	ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark	2
(ii)	**C** 9 minutes;		1
(iii)	In the following order only: • oxygen; • diffusion; • aerobic/cellular; • decreases/reduces/goes down;		4
(b) (i)	 must have 6 chromosomes shape not important	One mark for each correct cell drawn	2
(ii)	**B** metaphase; A is incorrect as chromosomes do not align at the equator during anaphase C is incorrect as chromosomes do not align at the equator during prophase D is incorrect as chromosomes do not align at the equator during telophase		1
			Total 10

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 5 of 7 · Question 2 part a.iii · 4 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

uestion number	Answer	Notes	Mark s

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 6 of 7 · Question 2 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
2	(a)	(i)	nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;;	one mark for two correct two marks for four correct	2
		(ii)	• large surface area; • thin epithelium/walls;	R cell wall/cell membrane	2
	(b)	(i)	B; 0.04% A, C and D are all incorrect figures		1
		(ii)	• 1200×16; • $\frac{19200}{1000}$ = 19 dm^3;	ECF if any value used from table Accept 19.2	2
		(iii)	• $\frac{20 \times 10\,500;}{100}$ • $\frac{2100 \times 20;}{100}$ • = 420 cm^3;	2 mark ECF	3
		(iv)	• more carbon dioxide in blood/lungs/body; • detected by (chemo)receptors (in arteries); • rate of breathing/tidal volume increases; • increases to remove carbon dioxide;		4

Total 14 marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 6 of 7 · Question 2 part a.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how alveoli are adapted for gas exchange and effects of damage

Answer 7 of 7 · Question 4 part a.ii · 4 marks

2024 JUN · Paper 02 · find part a.ii in this mark scheme extract

Question number				
4	(a)	(i)	Correct labels to: trachea; bronchus/bronchi; alveoli;	(3)
		(ii)	• many alveoli/large surface area; • short diffusion distance/ alveoli 1 cell thick/capillary 1 cell thick/ thin wall; • moist; • (dense/extensive/large) network of capillaries;	(4)
	(b)	(i)	any four from • reference to use of a nose clip; • reference to soda lime (to absorb carbon dioxide) • reference to sterilising mouth piece; • breathe in and out(normally) through a mouthpiece(connected to spirometer); • pen goes up and down (on a rotating drum);\ • measure peak to trough/ highest to lowest/top to bottom/ from the trace;	(4)
		(ii)	4600 ÷ 30 (x 100) = 1380; 4600 – 1380 = 3220 (cm^3); or 4600×0.7=3220 (cm^3);;	ecf: allow correct calculation from values given for mp1. Max 2 marks (2)

Total for Question 4 = 13 marks